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The benchmark inclusion subsidy

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1 Big picture

- **Question.** Does benchmarking by asset managers affect real corporate decisions such as investment, mergers, spinoffs, and IPOs?
- **Core answer.** Yes. Benchmarking creates extra, partly inelastic demand for firms inside the benchmark. This makes risky cash flows more valuable when they are attached to a benchmark firm.
- **Main concept.** The paper calls this wedge the **benchmark inclusion subsidy**: a firm inside the benchmark values the same risky project more than an otherwise identical firm outside the benchmark.
- **Why the textbook result fails.** Standard capital budgeting says that a project’s value is independent of the identity of the firm considering it. Here, the identity matters because benchmark membership changes investor demand for claims on the project’s cash flows.
- **Mechanism.** Portfolio managers are paid partly on performance relative to a benchmark. To hedge underperformance relative to that benchmark, they hold some benchmark stocks regardless of their cash-flow variance.
- **Investment implication.** Benchmark firms accept projects with lower mean cash flows, higher variance, or higher positive correlation with their existing assets than non-benchmark twins would accept.
- **M&A implication.** A benchmark acquirer can create value by buying a non-benchmark target even without operating synergies, because the target’s cash flows become attached to a benchmark stock.
- **Spinoff implication.** If a benchmark firm spins off an asset and the new firm is outside the benchmark, the spun-off asset loses the subsidy and can fall in value.
- **IPO implication.** A company that goes public and joins a benchmark should be worth more than an otherwise similar company that goes public outside the benchmark.
- **ESG implication.** Benchmark design matters. Excluding “brown” firms from a benchmark removes their subsidy, while including “green” firms subsidizes them and can encourage other firms to mimic them.
- **Quantification.** The paper’s calibration, using S&P 500 institutional ownership of 83% and overall market institutional ownership of 67%, implies a cost-of-capital wedge of about **94 basis points** for a project undertaken by a benchmark firm versus its non-benchmark twin.
- **Economic takeaway.** Benchmarking is not only an asset-pricing phenomenon. It can change corporate finance by altering the effective discount rate used for projects, acquisitions, spinoffs, and public listings.

2 Data, objects, and measurement

2.1 What is measured in the model

The paper is primarily theoretical, but it uses empirical inputs to quantify the subsidy. The important objects are:

- benchmark membership, represented by an indicator vector $\mathbf{1}_b$;
- the fraction of investors who are direct investors, λ_D ;
- the fraction of investors who use portfolio managers, λ_M ;
- the manager compensation parameters a and b ;
- the variance and covariance of cash flows;
- the inelastic demand generated by benchmarking;
- institutional ownership, used to calibrate the strength of inelastic benchmark demand.

Interpretation: The central empirical object is not an accounting variable but a demand wedge: how much extra demand benchmarked managers create for securities inside the benchmark.

2.2 Benchmarking and manager compensation

A portfolio manager's compensation is assumed to have three parts:

$$w = ar_x + b(r_x - r_b) + c = (a + b)r_x - br_b + c, \quad (1)$$

where r_x is portfolio performance and r_b is benchmark performance.

- a rewards absolute performance.
- b rewards performance relative to the benchmark.
- $b > 0$ is the key assumption. It makes benchmark underperformance costly.
- The manager therefore wants to hold the benchmark partly as a hedge.

Interpretation: Benchmarking turns the benchmark into a hedging portfolio for the manager. This is why benchmark stocks receive demand that is partly independent of their variance.

2.3 Institutional ownership calibration

The quantitative section uses institutional ownership from FactSet/LionShares for 2000–2017.

- Stocks in the S&P 500 have institutional ownership of about 83%.
- The overall market has institutional ownership of about 67%.
- The difference is interpreted as evidence of extra inelastic demand for benchmark stocks.
- The baseline calibration treats the S&P 500 as the benchmark and represents the market with ten size portfolios based on NYSE size deciles.
- The three largest deciles are treated as being inside the benchmark.

Interpretation: The paper does not need to claim that all institutional ownership is passive. The model's key input is that benchmark membership changes demand in a way that is less sensitive to project risk.

2.4 Cash-flow heterogeneity

To quantify the subsidy, the paper specifies ten asset cash-flow processes:

$$D_i = \alpha_i^0 [\bar{\mu} + \alpha_i^1 Z + (1 - \alpha_i^1) \varepsilon_i], \quad i = 1, \dots, 10. \quad (2)$$

- α_i^0 captures size.
- α_i^1 governs exposure to systematic versus idiosyncratic risk.
- Z is a common factor.
- ε_i is an idiosyncratic shock.

Interpretation: Bigger firms have larger cash-flow volatility in dollar terms. Because the subsidy rises with cash-flow risk, larger benchmark firms can receive a larger absolute subsidy.

2.5 Empirical evidence reviewed by the paper

The paper does not present a new large empirical test of corporate decisions. Instead, it connects the model to existing empirical work:

- index additions and deletions raise or lower stock prices;
- sin stocks can be in an index but excluded from investor benchmarks, helping separate index membership from benchmark demand;
- S&P 500 and MSCI index inclusion are associated with more investment, equity issuance, R&D, employment, and acquisitions;
- Russell 1000/2000 cutoff evidence shows that benchmark demand can move prices even for firms with similar size and liquidity;
- benchmarking intensity is associated with lower long-run returns, consistent with price pressure from benchmark demand.

Interpretation: The empirical evidence is indirect but broad. The distinctive predictions that remain most open are about how benchmark status changes IPO incentives and the choice of merger targets.

3 Models and theoretical specifications

3.1 Baseline economy without portfolio managers

There are two dates, three risky assets $i \in \{1, 2, y\}$, and a risk-free bond. Risky cash flows are

$$D_i \sim N(\mu_i, \sigma_i^2), \quad i = 1, 2, y, \quad (3)$$

with uncorrelated cash flows and unit supply. Direct investors have CARA utility,

$$U(W) = -e^{-\gamma W}. \quad (4)$$

Their demand for risky asset i is

$$x_i = \frac{\mu_i - S_i}{\gamma \sigma_i^2}. \quad (5)$$

Market clearing gives

$$S_i = \mu_i - \gamma \sigma_i^2. \quad (6)$$

If firm i merges with firm y , the combined price is

$$S'_i = \mu_i + m u_y - \gamma(\sigma_i^2 + \sigma_y^2) = S_i + S_y. \quad (7)$$

Interpretation: Without benchmarking, combining the cash flows of two firms creates no pure valuation gain. Ownership identity does not matter.

3.2 Adding portfolio managers

Let direct investors have population share λ_D , portfolio managers have share λ_M , and fund investors have share λ_F . The benchmark in the simple example is firm 1.

A direct investor's demand remains

$$x_i^D = \frac{\mu_i - S_i}{\gamma \sigma_i^2}. \quad (8)$$

The manager's demand for benchmark stock 1 is

$$x_1^M = \frac{1}{a+b} \frac{\mu_1 - S_1}{\gamma \sigma_1^2} + \frac{b}{a+b}, \quad (9)$$

while for non-benchmark stocks,

$$x_i^M = \frac{1}{a+b} \frac{\mu_i - S_i}{\gamma \sigma_i^2}, \quad i = 2, y. \quad (10)$$

Interpretation: The manager holds a scaled mean-variance portfolio plus a fixed benchmark hedge. The second term, $b/(a+b)$, is the source of the inelastic demand.

Define

$$\Lambda = \left[\frac{\lambda_M}{a+b} + \lambda_D \right]^{-1}. \quad (11)$$

The simple equilibrium prices are

$$S_1 = \mu_1 - \gamma \Lambda \sigma_1^2 \left(1 - \lambda_M \frac{b}{a+b} \right), \quad (12)$$

$$S_2 = \mu_2 - \gamma \Lambda \sigma_2^2, \quad S_y = \mu_y - \gamma \Lambda \sigma_y^2. \quad (13)$$

Interpretation: If firms 1 and 2 have identical cash flows, firm 1 is worth more because benchmarked managers reduce the effective supply of its shares.

3.3 Simple merger subsidy

If non-benchmark firm 2 acquires y , the combined value remains additive:

$$S'_2 = S_2 + S_y. \quad (14)$$

If benchmark firm 1 acquires y , the new price is

$$S'_1 = S_1 + S_y + \gamma \Lambda \sigma_y^2 \lambda_M \frac{b}{a+b}. \quad (15)$$

The additional term is the simple benchmark inclusion subsidy.

Interpretation: A benchmark acquirer creates financial value by moving the target cash flows into the benchmark. The gain is proportional to the target's cash-flow variance.

3.4 Index effect in the simple example

Let S'_y denote the price of y if it were inside the benchmark. Then

$$S'_y - S_y = \gamma \Lambda \sigma_y^2 \lambda_M \frac{b}{a+b}, \quad (16)$$

and the index effect is

$$\frac{S'_y - S_y}{S_y} = \frac{\gamma \Lambda \sigma_y^2}{S_y} \lambda_M \frac{b}{a+b}. \quad (17)$$

Interpretation: The model predicts that the index effect is larger for firms with riskier cash flows, even after controlling for the pre-inclusion stock price.

3.5 General model with many correlated firms

There are n risky stocks with jointly normal cash flows,

$$D = (D_1, \dots, D_n)' \sim N(\mu, \Sigma). \quad (18)$$

The first k stocks are in the benchmark. The benchmark indicator is

$$\mathbf{1}_b = (\underbrace{1, \dots, 1}_k, \underbrace{0, \dots, 0}_{n-k})'. \quad (19)$$

Direct investors demand

$$x^D = \frac{1}{\gamma} \Sigma^{-1} (\mu - S), \quad (20)$$

while managers demand

$$x^M = \frac{1}{a+b} \frac{1}{\gamma} \Sigma^{-1} (\mu - S) + \frac{b}{a+b} \mathbf{1}_b. \quad (21)$$

Equilibrium prices are

$$S = \mu - \gamma \Lambda \Sigma \left(\mathbf{1} - \lambda_M \frac{b}{a+b} \mathbf{1}_b \right). \quad (22)$$

Interpretation: Expected returns depend on both the market and the benchmark. The model generates a two-factor CAPM rather than the standard one-factor CAPM.

3.6 Project investment in the general model

Suppose firm i considers a project with cash flow

$$D_y \sim N(\mu_y, \sigma_y^2), \quad \text{Corr}(D_i, D_y) = \rho_{iy}. \quad (23)$$

The project costs I . If firm i invests, its cash flows become $D_i + D_y$.

The change in the investing firm's stockholder value is

$$\Delta S_i = \mu_y - I - \gamma \Lambda (\sigma_y^2 + \rho_{iy} \sigma_i \sigma_y) \left(1 - \lambda_M \frac{b}{a+b} \mathbf{1}_i \right) \quad (24)$$

$$- \gamma \Lambda \sum_{j=1}^n \rho_{jy} \sigma_j \sigma_y \left(1 - \lambda_M \frac{b}{a+b} \mathbf{1}_j \right). \quad (25)$$

Interpretation: The first terms are the project mean minus investment cost. The rest are risk penalties. Benchmark status changes the penalty applied to the project's own risk and its covariance with the investing firm's assets.

3.7 Benchmark inclusion subsidy

Consider two otherwise identical firms, one inside and one outside the benchmark, with $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$ and $\rho_{\text{in},y} = \rho_{\text{out},y} = \rho_y$. The difference in incremental stockholder value is

$$\Delta S_{\text{in}} - \Delta S_{\text{out}} = \gamma \Lambda (\sigma_y^2 + \rho_y \sigma \sigma_y) \lambda_M \frac{b}{a+b}. \quad (26)$$

Assumption 1 is

$$\sigma_y^2 + \rho_y \sigma \sigma_y > 0. \quad (27)$$

Interpretation: If Assumption 1 holds, the benchmark firm values the project more. The first term is the index-effect component. The covariance term is new: projects positively correlated with the firm's existing assets receive a larger subsidy.

3.8 Project valuation proposition

Proposition 1. A firm in the benchmark is more likely to invest in a project than a firm outside the benchmark if and only if Assumption 1 holds. More precisely, all else equal, a benchmark firm accepts projects with lower μ_y , larger σ_y^2 , and/or larger ρ_y than an otherwise identical firm outside the benchmark.

Interpretation: The proposition overturns standard capital budgeting independence. The project-specific discount rate differs across firms because investor demand differs across firms.

3.9 Risk-free and hedging projects

If $\sigma_y^2 = 0$, the project is risk free and all firms value it identically.

If the project is sufficiently negatively correlated with the firm's assets, then

$$\rho_y \leq -\frac{\sigma_y}{\sigma}. \quad (28)$$

In this case the project hedges cash-flow risk. A benchmark firm values the hedge less because its cash-flow risk is already subsidized by benchmark demand.

Interpretation: Benchmarking subsidizes risk, not safety. This is why the subsidy is largest for risky projects and vanishes for risk-free projects.

3.10 Cost of capital wedge

Define the project-specific cost of capital r_i by

$$\Delta S_i = -I + \frac{\mu_y}{1 + r_i}. \quad (29)$$

Then the difference between non-benchmark and benchmark project cost of capital is

$$r_{out} - r_{in} = \frac{\mu_y}{\Delta S_{out} + I} - \frac{\mu_y}{\Delta S_{in} + I} = \frac{\mu_y(\Delta S_{in} - \Delta S_{out})}{(\Delta S_{in} + I)(\Delta S_{out} + I)}. \quad (30)$$

Interpretation: The dollar subsidy maps directly into a lower project cost of capital for a benchmark firm.

3.11 Mergers, acquisitions, IPOs, and spinoffs

- **M&A.** A benchmark firm is more likely to acquire an outside target than a non-benchmark firm is, especially when the target's cash flows are risky and positively correlated with the acquirer's cash flows.
- **IPO.** A firm joining the benchmark at IPO receives an inclusion value. If another firm is pushed out, the IPO value depends on both the entrant's risk and the covariance with the displaced firm.
- **Spinoff.** If a benchmark firm spins off a division and the new company is not in the benchmark, the division loses the subsidy.

Interpretation: These applications show why benchmarking can affect corporate transactions even without operating synergies.

3.12 Passive asset management

A limiting case has passive managers with $b \rightarrow \infty$, so managers hold only the benchmark portfolio. If active and passive managers coexist, prices become

$$S = \mu - \gamma \Lambda \Sigma \left[\mathbf{1} - \left(\lambda_M^A \frac{b}{a+b} + \lambda_M^P \right) \mathbf{1}_b \right]. \quad (31)$$

The subsidy becomes

$$\gamma \Lambda (\sigma_y^2 + \rho_y \sigma \sigma_y) \left(\lambda_M^A \frac{b}{a+b} + \lambda_M^P \right). \quad (32)$$

Interpretation: Passive managers hold the benchmark irrespective of characteristics, so the subsidy rises as passive management becomes more important.

4 Identification and conceptual design

4.1 What identifies the theoretical effect

- The central comparison is between two otherwise identical firms, one inside and one outside the benchmark.
- The only difference is the investor-demand schedule induced by benchmarked managers.
- The benchmark firm receives extra demand for its risky cash flows.
- Therefore, the same project generates a larger stockholder-value increase inside the benchmark.

Interpretation: The model is a clean theoretical identification argument: hold cash flows fixed and change benchmark status.

4.2 Why the manager contract matters

- If $b = 0$, there is no relative-performance component.
- Without relative performance, managers have no benchmark hedge demand.
- Then benchmark and non-benchmark firms are priced symmetrically.
- Therefore the benchmark inclusion subsidy depends on the prevalence and strength of benchmark-based evaluation.

Interpretation: The mechanism is not generic institutional ownership. It is benchmark-linked ownership.

4.3 Why the covariance term matters

- The subsidy rises with project variance.
- The subsidy also rises when the project is positively correlated with the acquiring or investing benchmark firm's existing cash flows.
- If the project is a clone of existing assets, the covariance subsidy is large.
- If the project is a hedge, the benchmark firm values it less than a non-benchmark firm.

Interpretation: This is a sharp cross-sectional prediction. Benchmark firms should especially like risky projects that resemble their existing assets.

4.4 What makes the calibration informative

- The model is not estimated using a full structural empirical exercise.
- It calibrates the strength of inelastic benchmark demand using the institutional ownership gap between S&P 500 stocks and the market.
- It uses size portfolios and market betas to discipline cash-flow heterogeneity.
- The 94 bps number is therefore an indicative magnitude, not a definitive estimate.

Interpretation: The calibration is useful because it shows that the model’s effect is large enough to matter for real decisions.

4.5 What the paper cannot fully prove

- The quantitative exercise relies on CARA-normal assumptions.
- Institutional ownership is an imperfect measure of inelastic benchmark demand.
- Benchmark membership can be endogenous to firm size, liquidity, and prospects.
- Existing empirical evidence is supportive but mostly indirect.
- The paper does not provide a new causal empirical test of all corporate action predictions.
- Welfare effects are ambiguous because risk is reallocated across direct investors, managers, and fund investors.

Interpretation: The paper’s strongest contribution is the mechanism and its testable cross-sectional predictions, not a final empirical measurement of the subsidy in every setting.

5 Tables, figures, and original excerpts

This section keeps the same *Read* plus *Economic message* format, and groups adjacent original-paper excerpts to save space. All images are directly cropped from the paper.

5.1 Original excerpt: manager demand and inelastic benchmark demand

Read:

- The direct investor’s demand is the standard CARA-normal mean-variance demand.
- The manager’s demand for benchmark stock contains two pieces: a scaled mean-variance term and a fixed benchmark term $b/(a + b)$.
- The fixed term is independent of price, expected cash flow, and variance.
- This term is what makes benchmark stocks special.

Economic message: The model begins with a simple but powerful idea: benchmarked managers buy part of the benchmark because their compensation penalizes benchmark underperformance.

S_1) and $r_b = D_1 - S_1$. For simplicity, we assume that a , b , and c are set exogenously.⁸

A direct investor's demand for asset i continues to be

$$x_i^D = \frac{\mu_i - S_i}{\gamma \sigma_i^2}, \quad i = 1, 2, y. \quad (6)$$

A manager's demands for stocks inside and outside the benchmark are

$$x_1^M = \frac{1}{a+b} \frac{\mu_1 - S_1}{\gamma \sigma_1^2} + \frac{b}{a+b}, \quad (7)$$

$$x_i^M = \frac{1}{a+b} \frac{\mu_i - S_i}{\gamma \sigma_i^2}, \quad i = 2, y. \quad (8)$$

Managers' portfolio choices differ from those of the direct investors in two ways. First, they hold a scaled version of the same mean-variance portfolio as the one held by the direct investors. The reason for the scaling is that, as we can see from the first term in (5), for each share that the manager holds, she gets a fraction $a+b$ of the total return. Thus the manager scales her asset holdings by $1/(a+b)$ relative to those of a direct investor.

Second, and more importantly, the portfolio managers are penalized by b for underperforming the benchmark. Because of this penalty, the manager always holds $b/(a+b)$ shares of stock 1 (or more generally whatever is in the benchmark), regardless of its risk-return characteristics. This consideration explains the second term in (7). This inelastic demand for the benchmark will be critical for all of

Figure 1: Original excerpt: manager demands and benchmark demand.

5.2 Original excerpt: simple merger subsidy

Read:

- If non-benchmark firm 2 acquires y , the combined value is additive: $S'_2 = S_2 + S_y$.
- If benchmark firm 1 acquires y , combined value exceeds additivity.
- The extra term is $\gamma \Lambda \sigma_y^2 \lambda_M b / (a+b)$.
- The term is larger when the target's cash-flow variance is larger.

Economic message: A benchmark acquirer can create pure financial synergy by moving target cash flows into the benchmark.

benchmark means that the adverse effects of variance that typically reduce the demand for any stock, are less relevant for the assets in the benchmark.⁹

Next, consider potential mergers. Suppose first that firm y is acquired by the non-benchmark firm (firm 2). The new demands of direct investors and portfolio managers for the stock of firm 2 are

$$x_2^D = \frac{\mu_2 + \mu_y - S_2'}{\gamma(\sigma_2^2 + \sigma_y^2)}, \quad (12)$$

$$x_2^M = \frac{1}{a+b} \frac{\mu_2 + \mu_y - S_2'}{\gamma(\sigma_2^2 + \sigma_y^2)}. \quad (13)$$

The new equilibrium price of firm 2's stock is

$$S_2' = \mu_2 + \mu_y - \gamma \Lambda (\sigma_2^2 + \sigma_y^2) = S_2 + S_y. \quad (14)$$

As before, the combined value of firm 2, continues to be the sum of the initial value plus the value of y .

Suppose instead that y is acquired by firm 1, which is in the benchmark. Re-normalizing the combined number of shares of firm 1 to one, the demands for the stock of the combined firm are

$$x_1^D = \frac{\mu_1 + \mu_y - S_1'}{\gamma(\sigma_1^2 + \sigma_y^2)}, \quad (15)$$

$$x_1^M = \frac{1}{a+b} \frac{\mu_1 + \mu_y - S_1'}{\gamma(\sigma_1^2 + \sigma_y^2)} + \frac{b}{a+b}. \quad (16)$$

Our next major finding is that there is a benchmark inclusion subsidy. Specifically, the new price of firm 1's shares is

$$S_1' = \mu_1 + \mu_y - \gamma \Lambda (\sigma_1^2 + \sigma_y^2) = S_1 + S_y.$$

ket penalizes the variance of y 's cash flows less when they are inside firm 1 rather than firm 2.

In contrast, notice that if firm y had started out inside the benchmark, then S_1' would be exactly equal to the sum of prices of stocks 1 and y . In that case, the inelastic demand for the stock would already have been embedded in its price before the merger. So, the extra value of acquisition that accrues to firm 1 relative to firm 2 arises from the increase in the price of y when it becomes part of the benchmark.

To put this more formally, let S_y' denote the price of y if it were inside the benchmark. Then $S_y' - S_y = \gamma \Lambda \sigma_y^2 \lambda_M b / (a+b)$, which is precisely the extra term in Eq. (17). Notice that this is directly related to the "index effect" estimated in the literature, which is the percentage change in a firm's stock price when it joins the benchmark, and in our model is

$$\frac{S_y' - S_y}{S_y} = \frac{\gamma \Lambda \sigma_y^2}{S_y} \lambda_M \frac{b}{a+b}, \quad (18)$$

where S_y is given by (11).

Thus, in this simple example, the benchmark inclusion subsidy reduces to the index effect for the target firm. As we will show in the general model in Section 4, if we allow for any correlation between the acquirer's and the target's cash flows, the benchmark inclusion subsidy will have an additional term accounting for the correlation. When the correlation is positive, the subsidy exceeds the index effect for the target firm.

It is worth noting that our model predicts that the index effect is larger for firms with riskier cash flows. This can be seen from Eq. (18), where the index effect is increasing in σ_y^2 , even after controlling for the stock price before the inclusion. The literature so far has focused on estimating the average index effect. In contrast, our model

Figure 2: Original excerpt: merger example and simple benchmark inclusion subsidy.

5.3 Original excerpts: general project valuation and Proposition 1

Read:

- The general model allows cash-flow correlations across firms and projects.
- Equation (24) gives the stockholder-value change from investment.
- Equation (25) gives the closed-form benchmark inclusion subsidy.
- Proposition 1 states the investment rule: benchmark firms invest more when $\sigma_y^2 + \rho_y \sigma \sigma_y > 0$.

Economic message: The closed-form subsidy shows exactly which project characteristics matter: variance, covariance with existing assets, the size of benchmarked management, and the strength of relative-performance incentives.

it, where $\Delta S_i^{(0)} = I$. If firm i is in the benchmark, then the additional shares enter the benchmark.

To proceed, suppose firm i (and only firm i) invests in the project. Then the new cash flows are $D^{(0)} = D + (0, \dots, 0, D_y, 0, \dots, 0)^T$, distributed according to

$N(\mu^{(0)}, \Sigma^{(0)})$, where $\mu^{(0)} = \mu + (0, \dots, 0, \mu_y, 0, \dots, 0)^T$

and

$$\Sigma^{(0)} = \Sigma + \begin{pmatrix} 0 & \rho_{iy}\sigma_i\sigma_y & 0 & \dots & 0 \\ \rho_{iy}\sigma_i\sigma_y & \dots & \sigma_y^2 + 2\rho_{iy}\sigma_i\sigma_y & \dots & \rho_{iy}\sigma_i\sigma_y \\ 0 & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & 0 \end{pmatrix} \quad (22)$$

Denote $I^{(0)} = (0, \dots, 0, I, 0, \dots, 0)^T$.

Lemma 3 (Post-Investment Asset Prices). The equilibrium stock prices when firm i invests in the project are given by

$$S^{(0)} = \mu^{(0)} - I^{(0)} - \gamma \Delta \Sigma^{(0)} \left(1 - \lambda_M \frac{b}{a+b} \right) \mathbf{b}. \quad (23)$$

Section 3, applies to all firms contemplating investment, including the non-benchmark ones. We comment further on how it affects firms' investment incentives at the end of this subsection.

We are now ready to derive the benchmark inclusion subsidy in this generalized setting. Consider incentives of firm i to invest in a project. It will do so if its stockholder value goes up as a result of the investment, that is, if $\Delta S_i > 0$. Consider two firms, i_{in} and i_{out} , one in the benchmark and the other is not (i.e., $i_{in} \leq k$ and $i_{out} > k$). Suppose that their cash flows with and without the project are identical; specifically, $\sigma_{i_{in}} = \sigma_{i_{out}} = \sigma$ and $\rho_{iy} = \rho_{i_{out}y} = \rho_y$. The difference in the incremental stockholder value created by the investment for the two firms is

$$\Delta S_{i_{in}} - \Delta S_{i_{out}} = \gamma \Delta (\sigma_y^2 + \rho_y \sigma_i \sigma_y) \lambda_M \frac{b}{a+b}. \quad (25)$$

The right hand side of (25) is the analytical expression for the benchmark inclusion subsidy.

Assumption 1. $\sigma_y^2 + \rho_y \sigma_i \sigma_y > 0$.

So long as Assumption 1 holds, the benchmark inclusion subsidy is positive, and the increase in the stockholder value for the firm in the benchmark is larger than that for the firm outside the benchmark.

That is, projects with a positive covariance with the benchmark are more valuable, even if a non-benchmark firm undertakes them. This is because prices of benchmark stocks are inflated by the inelastic demand from portfolio managers, which leads direct investors (and managers through their mean-variance portfolios) to substitute into assets that provide exposure to the benchmark without being in the benchmark itself. Consequently, of the two covariances that enter the extra variance, one is subsidized regardless of which firm invests and the other is subsidized only when the investing firm is a benchmark firm. Hence, one of the two covariances drops out from the difference-in-differences.

The presence of the benchmark inclusion subsidy translates into different investment rules for firms inside and outside the benchmark. We formalize this result in Proposition 1.

Proposition 1 (Project Valuation). A firm in the benchmark is more likely to invest in a project than a firm outside the benchmark if and only if Assumption 1 holds. More precisely, all else equal, a firm in the benchmark accepts projects with a lower mean μ_y , larger variance σ_y^2 , and/or larger correlation ρ_y than an otherwise identical firm outside the benchmark if and only if Assumption 1 holds.

Proposition 1 is at odds with the textbook treatment of valuation taught in basic corporate finance courses. The usual rule states that a project's value is independent of which firm undertakes it and is simply given by the

quantity, a benchmark firm will value such a project less than a non-benchmark firm would.

Finally, as we mentioned earlier, our model also implies that projects correlated with assets inside and outside the benchmark are valued differently (by firms both inside and outside the benchmark). Projects that are positively correlated with the benchmark provide alternative cheaper exposure to the benchmark firms' cash flows. This is reflected in Eq. (24), which shows that for any firm, investing in a project that is positively correlated with a component of the benchmark is more beneficial than if the project has the same degree of correlation with an asset outside of the benchmark.

To illustrate this insight graphically, Fig. 1 plots the change in the stockholder value ΔS_i given by (24) as a function of ρ_{iy} , where the solid line corresponds to some asset j inside the benchmark, and the dashed line to some j outside the benchmark. In the figure, for concreteness, the investing firm i is in the benchmark. If i were outside the benchmark, these lines would shift down in parallel.

The figure shows that the change in stockholder value, ΔS_i , is decreasing in the correlation coefficient ρ_{iy} . If j is in the benchmark, then the downward sloping line is flatter. Moreover, the solid line is above the dashed line for positive correlations and below for negative correlations. This is because positive (negative) correlation of the project with an asset in the benchmark is penalized (rewarded) less than the same correlation with an asset outside the benchmark.

Figure 3: Original excerpt: general project value and subsidy formula.

Figure 4: Original excerpt: Proposition 1 and intuition.

5.4 Figure 1 and cost-of-capital/M&A excerpt

Read:

- Figure 1 plots the stockholder-value change as a function of correlation with another asset.
- The solid line corresponds to correlation with an asset inside the benchmark; the dashed line corresponds to correlation with an asset outside the benchmark.
- Positive correlation with the benchmark is rewarded more.
- The cost-of-capital excerpt translates the dollar subsidy into a percentage discount-rate wedge.
- Proposition 2 applies the logic to mergers and acquisitions.

Economic message: Benchmarking changes not just the value of benchmark firms but also the value of correlation with benchmark cash-flow risk.

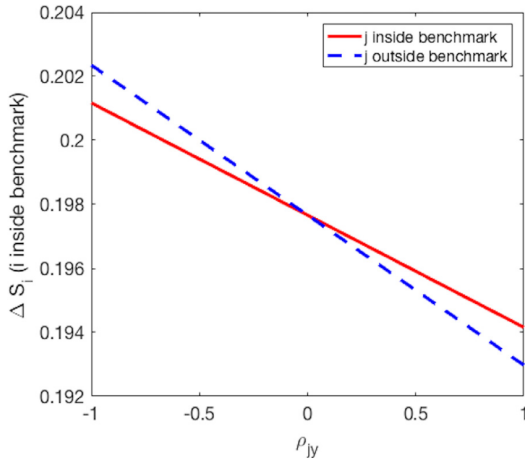


Fig. 1. Change in the stockholder value, ΔS_i , as a function of correlations of project y 's cash flows with cash flows of assets inside and outside the benchmark, ρ_{iy} for some $j \leq k$ and some $j > k$. Parameter values: $n = 5$, $k = 3$, $\mu_y = 1.2$, $I = 1$, $\sigma_j = 0.15$, $\sigma_y = 0.1$, $\rho_{iy} = 0$ unless it is plotted on the horizontal axis, $\rho_{j\ell} = 0$, $\ell \neq j$, $j = 1, \dots, n$, $\gamma = 2$, $\lambda_M = 0.3$, $a = 0.008$, $b = 0.042$. Investing firm: $i = 1$. Solid line: $j = 2$. Dashed line: $j = 4$.

Figure 5: Original Figure 1: stockholder-value change and project correlation.

4.1.1. Cost of capital

Notice that the benchmark inclusion subsidy is directly related to the project-specific cost of capital for an investment. Define firm i 's cost of capital, applied to the investment project y , r_i , by $\Delta S_i = -I + \mu_y / (1 + r_i)$, where ΔS_i , defined in (24), measures the increase in stockholder value from adopting the project. Then the difference in the cost of capital for firms outside and inside the benchmark is

$$r_{out} - r_{in} = \frac{\mu_y}{\Delta S_{out} + I} - \frac{\mu_y}{\Delta S_{in} + I} = \frac{\mu_y (\Delta S_{in} - \Delta S_{out})}{(\Delta S_{in} + I)(\Delta S_{out} + I)}. \quad (26)$$

Notice that unlike (25), which is measured in dollars, (26) is unit-free (or can be expressed in percent). For that reason, we will use (26) for the quantitative analysis in Section 5.

4.2. Mergers and acquisitions

As we have seen in the example considered in Section 3, the model can also be used to think about mergers and acquisitions.

Proposition 2 (Mergers and Acquisitions). Suppose firm i considers acquiring firm y that is outside the benchmark, and

Figure 6: Original excerpt: cost of capital and M&A proposition.

5.5 Passive management and Table 1: calibration inputs

Read:

- The passive-management proposition says the subsidy is larger when the passive share of asset management is larger.
- Table 1 lists the calibration parameters.
- $\gamma\Lambda$ is chosen to match the market risk premium.
- σ_z^2 is chosen to match market return volatility.
- $\lambda_M/(a+b)$ and $b\zeta$ are chosen to match institutional ownership in the market and in the benchmark.

Economic message: The calibration is disciplined by market size, beta, volatility, and ownership data. The key quantitative input is the gap between benchmark and market institutional ownership.

As we mentioned earlier, a limiting case of our setup when $b \rightarrow \infty$ corresponds to having passive asset management. In this case, it is easy to see that managers hold only the benchmark portfolio, i.e. $x^M = \mathbf{1}_b$. A generalization of our model would be to include both active and passive managers. If we denote the fractions of them in the economy by λ_M^A and λ_M^P , then the equilibrium stock prices would be

$$S = \mu - \gamma\Lambda\Sigma\left(\mathbf{1} - \left[\lambda_M^A \frac{b}{a+b} + \lambda_M^P\right]\mathbf{1}_b\right), \quad (28)$$

where $\Lambda = [\lambda_M^A/(a+b) + \lambda_D]^{-1}$.

All of our results extend to this case. Passive managers hold benchmark stocks irrespective of their characteristics, and they invest nothing in the mean-variance portfolio. Therefore, with passive managers the benchmark inclusion subsidy becomes even larger.¹⁹ In particular, instead of the expression in (25), the subsidy becomes $\gamma\Lambda(\sigma_z^2 + \rho_\gamma\sigma_\gamma)[\lambda_M^A b/(a+b) + \lambda_M^P]$. The following proposition shows that this value is increasing if more managers are passive.

Proposition 4 (Passive Asset Management). The benchmark inclusion subsidy is larger when λ_M^P/λ_M^A is larger, holding $\lambda_M^P + \lambda_M^A$ constant.

An analogous comparative-statics statement applies to

Table 1
Parameter values.

Parameter	Value	Explanation
α^0	(1, 0.1936, 0.1024, 0.0605, 0.0382, 0.0251, 0.0165, 0.0104, 0.0053, 0.0011)	For the largest asset, α_1^0 is normalized to be 1; the other 9 are chosen to match data for estimates of size ratios of deciles 2–10 relative to decile 1: (0.1914, 0.1008, 0.0594, 0.0373, 0.0245, 0.0161, 0.0101, 0.0051, 0.0011)
α^1	(0.1, 0.1174, 0.1224, 0.1269, 0.1317, 0.1342, 0.1386, 0.1442, 0.1510, 0.1504)	α_1^1 is set to be 0.1; the other 9 are chosen to match estimates of market betas for deciles 2–10: (1.0914, 1.1407, 1.1852, 1.2339, 1.2600, 1.3047, 1.3634, 1.4344, 1.4280)
$\gamma\Lambda$	0.2789	Chosen to match the market risk premium of 7.7%
σ_1^2	316.33	Chosen to match the standard deviation of return on the market of 19.7%
σ_z^2	0.0957	Chosen to match the average idiosyncratic risk of the average asset (across the size portfolios), 3.5%
$\lambda_M/(a+b)$	0.302	Chosen to match the equally-weighted institutional ownership of all stocks in the market, 67%
$b\zeta$	1.9	Chosen to match the equally-weighted institutional ownership of benchmark stocks, 83%

Figure 7: Original excerpt: passive asset management.

Table 1: Original Table 1: parameter values.

5.6 Table 2: benchmark inclusion subsidy and institutional ownership

Read:

- The table reports the difference in cost of capital for otherwise identical inside and outside firms, in basis points.
- The baseline is the center cell: benchmark institutional ownership 83%, market institutional ownership 67%, and a subsidy of 94 bps.
- If benchmark and market institutional ownership are equal, the subsidy is zero.
- If benchmark ownership is 91% and market ownership is 59%, the subsidy rises to 260 bps.
- If benchmark ownership is 75% and market ownership is 75%, the subsidy is zero.

Economic message: The subsidy is governed by the ownership gap: the larger the extra inelastic demand for benchmark stocks, the larger the cost-of-capital advantage.

Table 2

The benchmark inclusion subsidy and institutional ownership. This table presents a sensitivity analysis of the difference in the cost of capital for new investments (in basis points) for identical firms outside vs. inside the benchmark, as institutional ownership varies. Each of the identical firms has the characteristics of asset 3. The top 3 size deciles of stocks are in the benchmark.

		Institutional ownership of the market		
		59%	67%	75%
Institutional ownership of the benchmark	75%	67	35	0
	83%	133	94	51
	91%	260	215	159

Table 2: Original Table 2: subsidy sensitivity to institutional ownership.

6 Short summary

- The paper studies the real effects of evaluating asset managers relative to benchmarks.
- Relative-performance contracts induce managers to hold benchmark stocks as a hedge against benchmark underperformance.
- This creates inelastic demand for benchmark securities.
- The demand makes risky cash flows more valuable inside the benchmark than outside.
- The difference is the benchmark inclusion subsidy.
- In a simple merger example, a benchmark firm creates value by acquiring a non-benchmark target even when cash flows do not change.
- In the general model, the subsidy equals $\gamma\Lambda(\sigma_y^2 + \rho_y\sigma\sigma_y)\lambda_M b/(a+b)$.
- The subsidy rises with project variance, positive covariance with existing assets, the share of benchmarked managers, and the strength of relative-performance incentives.
- Risk-free projects receive no subsidy.
- Hedging projects may be less valuable to benchmark firms.
- The subsidy lowers the project-specific cost of capital for benchmark firms.
- The model applies to investment, M&A, spinoffs, IPOs, and ESG benchmark design.
- Passive management amplifies the subsidy because passive managers hold benchmark stocks regardless of characteristics.
- Welfare effects are ambiguous because the subsidy changes risk sharing across investor groups.
- The calibration suggests the subsidy can be quantitatively large: about 94 bps in the baseline S&P 500 case.
- Existing empirical evidence on index effects, sin stocks, benchmark ownership, investment, acquisitions, R&D, and Russell index cutoffs is broadly consistent with the model.
- The most direct future tests would examine IPO benchmark motives and merger target selection by benchmark firms.

7 Conclusion

7.1 Core conclusion

Kashyap, Kovrijnykh, Li, and Pavlova show that benchmarking creates an inclusion subsidy for firms inside a benchmark. Because portfolio managers are evaluated against benchmarks, they demand benchmark stocks partly independent of those stocks' risk characteristics. This lowers the effective penalty on risky projects undertaken by benchmark firms.

7.2 Economic intuition

A risky project normally lowers value through a risk penalty. But if the project is attached to a benchmark firm, some of the demand for the resulting cash flows comes from benchmarked managers who hold the benchmark for compensation reasons. The market therefore penalizes the added variance less. The same cash flows can be worth more inside the benchmark than outside.

7.3 Takeaway

The paper changes how to think about capital budgeting and corporate transactions in a benchmarked asset-management economy. The identity of the firm considering a project can matter. Benchmark membership can alter discount rates, acquisition valuations, IPO incentives, spinoff values, and ESG benchmark effects. Asset management therefore has real consequences, not only asset-pricing consequences.